

$\text{Rep}(G)_R$ - (cat. of smooth R - $G(F)$ -reps) Wenging Wei

$$R = \mathbb{Z}[\frac{1}{p}], \dots$$

Let $H \subset G(F)$ open compact subgroup.

$$\text{Hecke algebra } \mathcal{H}(G//H)_R = R[H \backslash G(F) / H]$$

$$\mathcal{Z}(G//H)_R = \text{centre of } \mathcal{H}(G//H)_R.$$

Recall the goal:

1. $\mathcal{Z}(G//H)_R$ is a f.g. R -algebra
2. $\mathcal{H}(G//H)_R$ f.g. $\mathcal{Z}(G//H)_R$ -mod.

$$\mathcal{Z}_G := \text{centre of } \text{Rep}(G)_R = \text{End}(\text{Id}).$$

In Tong's talk: $\forall$ f.g. $V \in \text{Rep}(G)_R$

- ① $\text{Im}(\mathcal{Z}_G \rightarrow \text{End}_R(V))$ is a f.g. R -algebra
- ② V is $\mathcal{Z}_G$ -admissible, i.e. $\forall H \subset G(F)$, V^H is a f.g. $\mathcal{Z}_G$ -module.

By some arguments, it suffices to show above for $V = i_p(w)$.

i_p = normalized parabolic induction.

By Fargues-Scholze, there is a ring homo.

$$\mathrm{Exc}(W_F, \check{G}) \longrightarrow \mathcal{Z}_G$$

which is compatible w/ parabolic induction.

$$\mathrm{Exc}(W_F, \check{G}) \longrightarrow \mathrm{End}_{\check{G}}(\mathrm{ip}(W))$$

$$\downarrow \quad \hookrightarrow \quad \uparrow$$

$$\mathrm{Exc}(W_F, \check{M}) \longrightarrow \mathrm{End}_{\check{M}}(W)$$

$$\mathrm{Exc}(W_F, \check{G}) = \varprojlim \mathrm{Exc}(W_F/P_F^e, \check{G})$$

$$\mathrm{Exc}(W_F, \check{G}) \longrightarrow \mathrm{End}_{\check{G}}(V) \quad \text{factors through some } \mathrm{Exc}(W_F/P_F^e, \check{G})$$

f.g.

Now suppose we have proven result for M , W is f.g. over $\mathrm{Exc}(W_F/P_F^e, \check{M})$

We know that parabolic induction preserves admissibility.

It suffices to show that $\mathrm{Exc}(W_F^\circ/P_F^e, \check{G}) \rightarrow \mathrm{Ext}(W_F^\circ/P_F^e, \check{M})$ is finite.

By def'n,

$$\mathrm{Ext}(W_F^\circ/P_F^e, \check{G}) \longrightarrow \mathcal{O}(\mathcal{Z}^1(W_F^\circ/P_F^e, \check{G}))^{\check{G}}$$

We know it is an isom. if $\otimes_{\mathbb{Z}} \mathbb{Q}$.

$$\text{Exc}(W_F^\circ / P_F^e, \check{G}) \rightarrow \mathcal{O}(Z^1(W_F^\circ / P_F^e, \check{G}))^{\check{G}}$$

↓

↓ (*)

$$\text{Exc}(W_F^\circ / P_F^e, \check{M}) \rightarrow \mathcal{O}(Z^1(W_F^\circ / P_F^e, \check{M}))^{\check{M}}$$

Main Th. (*) is a finite map.

Cor $\text{Exc}(W_F^\circ / P_F^e, \check{G})_{\text{red}}$

↓

$$\text{Exc}(W_F^\circ / P_F^e, \check{M})_{\text{red}}$$

finite.

??



Modul' of tame parameters

W_F - Weil gp

$$\begin{array}{c} \text{Gal } \bar{F} / F \\ | \\ I_F \\ | \\ P_F \\ | \\ * \end{array} \Bigg) \hat{\mathbb{Z}}^{(p)}$$

$$\begin{array}{c} \bar{F} \\ | \\ F^{\text{tame}} = F^{\text{un}}(\sqrt[p+n]{\pi_F}) \\ | \\ F^{\text{un}} = F(\{p^{n-1}, n \in \mathbb{Z}_{>0}\}) \\ | \\ F \end{array}$$

$$0 \rightarrow I_F \rightarrow W_F \rightarrow \hat{\mathbb{Z}} \rightarrow 0$$

$$\begin{array}{ccccccc} & & & \downarrow & & & \\ & & & I & & & \\ 0 & \rightarrow & I_F & \rightarrow & \text{Gal } \bar{F} / F & \rightarrow & \hat{\mathbb{Z}} \rightarrow 0 \end{array}$$

Choose a $\text{Fr} \in W_F$.

Choose a generator $s \in I_F$ of I_F / P_F .

$$\text{Fr} \cdot s \cdot \text{Fr}^{-1} = s^q.$$

Let $W_F^\circ = \langle Fr, s \rangle \propto P_F \leftarrow \text{discretized Weil group}$
 $\parallel$
 $Fr \mathbb{Z}_\alpha s \mathbb{Z}[\frac{1}{q}]$

We are interested in $\mathbb{Z}^1(W_F^\circ, \check{G})$

$$R \mapsto \left\{ \phi: W_F^\circ \rightarrow {}^L \check{G} \begin{matrix} \swarrow \text{discrete top.} \\ \parallel \\ \check{G} \rtimes W_F \end{matrix} \right\} \quad \phi \text{ cts}$$

incl. $\searrow$ $\downarrow$
 W_F

$$W_t^\circ = \langle Fr, s \rangle$$

$$\mathbb{Z}^1(W_t^\circ, \check{G}) = \{ \phi \in \mathbb{Z}^1(W_F^\circ, \check{G}) : \phi|_{P_F} = \text{Id} \}$$

Assumptions ① $W_F^\circ \curvearrowright \check{G}$, P_F acts trivially

$\check{G}$ is tamely ramified

② $W_F^\circ \curvearrowright \check{G}$ preserves a Borel pairs $(\check{B}, \check{T})$

$\mathbb{Z}^1(W_t^\circ, \check{G})$ is a closed subscheme of $\check{G}^2$, cutting out by the rel'n

$$Fr \cdot s \cdot Fr^{-1} = s^q$$

$$\mathbb{Z}^1(W_t^\circ, \check{G}) \hookrightarrow (\check{G} \times Fr) \times (\check{G} \rtimes s)$$

① (base change to $\forall L = \bar{L}$) $\mathbb{Z}^1(W_F^\circ, \check{G}) \rightarrow \check{G} \rtimes s$

the image only contains finitely many $\check{G}$ -conj. orbits.

$$\textcircled{2} \quad Z^1(W_t^0, \check{h}) // \check{h} \rightarrow \check{h} \rtimes \text{Fr} // \check{h} \quad \text{is finite}$$

$$\textcircled{1} \quad W \subset \mathcal{O}_h, \quad \check{h} = h L_h$$

$$L\varphi: W_t^0 \rightarrow Lh, \quad F = L\varphi(\text{Fr}), \quad S = L\varphi(s)$$

$$\text{Fr} \cdot S \cdot \text{Fr}^{-1} = S^q \Rightarrow F \cdot S \cdot F^{-1} = S^q$$

Weil gp $\Delta \check{h}$ factors through some finite group of order e .

$$F^e \cdot S^e \in \check{h}. \quad \underbrace{F^e \cdot S^e \cdot F^{-e}}_{h L_h} = S^{e \cdot q^e}$$

Let $\{\lambda_1, \dots, \lambda_n\}$ be eigenvalues of S^e . then

$$\{\lambda_1^{q^e}, \dots, \lambda_n^{q^e}\} = \{\lambda_1, \dots, \lambda_n\} \quad \text{iterating} \quad \lambda_i^{q^e} = \lambda_i$$

$$S = S_{ss} \cdot S_{un}, \quad S_{ss}^{e(q^e-1)} = 1 \quad \leftarrow \text{finitely many up to } \check{h}\text{-conj.}$$

$$S_{ss}, S_{un} \in C_{\check{h}}^{\vee}(S_{ss})^0 \leftarrow \text{reductive}$$

In conn'd reductive group, there are finite unipotent elements up to conj.

$\Rightarrow$ finitely many choices of S up to $\check{h}$ -conj.

$$\dim Z^1(W_t^0, \check{h}) = \dim \check{h}$$

$$Z^1(W_t^0, \check{h}) \rightarrow \underbrace{\check{h} \rtimes S}_{\frac{4}{3}}$$

$$Z^1(W_t^o, \check{h})_{\bar{z}} = \{ \varphi \in Z^1(W_t^o, \check{h}) : L\varphi(s) = \bar{z} \}$$

$$Z^1(W_t^o, \check{h})_{(\bar{z})} = \{ \varphi \in Z^1(W_t^o, \check{h}) : L\varphi(s) \check{h}\text{-conj. to } \bar{z} \}$$

It's easy to see that $Z^1(W_t^o, \check{h})_{\bar{z}}$ is a $\check{h}_{\bar{z}}$ -torsor

$$\check{h}_{\bar{z}} = \underline{(\check{h}(\bar{z}))} \quad (\text{this gp scheme may not be reduced in general})$$

$$\phi_1, \phi_2 \in Z^1(W_t^o, \check{h})_{\bar{z}}$$

$$\phi_i(Fr) \phi_i(s) \phi_i(Fr) = \phi_i(s)^2, \quad i=1,2$$

$$\phi_1(s) = \phi_2(s) \Rightarrow \phi_1(Fr)^{-1} \phi_2(Fr) \in C_{\check{h}}(\bar{z})$$

$$Z^1(W_t^o, \check{h})_{(\bar{z})} = \check{h} \times^{\check{h}_{\bar{z}}} Z^1(W_t^o, \check{h})_{\bar{z}}$$

↓

underlying reduced scheme is smooth and of dim $\check{h}$.

$$\textcircled{2} \quad Z^1(W_t^o, \check{h}) // \check{h} \longrightarrow \check{h} \rtimes Fr // \check{h} \quad \text{is finite}$$

Quick review on GIT quotient:

$$R, \quad X = \text{Spec } A \xrightarrow{\text{b.p.}} \text{Spec } R, \quad H\text{-reductive group } \curvearrowright X$$

$$X // H = \text{Spec } A^H$$

① A^H is a f.g. R -algebra

② $L = \bar{L}$, closed $H(L)$ -orbits of $X(L) \Leftrightarrow L$ -points of $X//H$

③ Hilbert - Mumford criterion.

$$x \in X(L)$$

$H \cdot x$ is closed $\Leftrightarrow \forall \gamma: \mathbb{G}_m \rightarrow H$, if $y = \lim_{t \rightarrow 0} \gamma(t) \cdot x$ exists
then $y \in Hx$.

$$Z^1(W_t^0, \check{G}) //_{\check{G}}^{\vee}$$

$\varphi: W_t^0 \rightarrow L_G$ L -point of $Z^1(W_t^0, \check{G})$

$\check{G} \cdot \varphi$ is closed $\Leftrightarrow \forall \gamma: \mathbb{G}_m \rightarrow H$, $\varphi' = \lim_{t \rightarrow 0} \gamma(t) \cdot \varphi$ exists,

$$\varphi' \sim_{\check{G}}^{\vee} \varphi$$

$\varphi' = \lim_{t \rightarrow 0} \gamma(t) \cdot \varphi$ exists $\Leftrightarrow \forall g \in W_t^0$, $\varphi'(g) = \lim_{t \rightarrow 0} \gamma(t) \varphi(g)$ exists

$\Leftrightarrow \text{Im}^L \varphi$ lies in the parabolic subgroup of L_G defined by γ

$\check{G} \cdot \varphi$ is closed $\Leftrightarrow \forall P \subset L_G$ if the image of ${}^L \varphi$ lies in P , then you

can $\check{G}$ -conj. it so that $\text{Im}^L \varphi$ lies in M .

Such reps are called semisimple reps.

$L\varphi: W_t^0 \rightarrow L_G$ is a ss rep.

$$F = L\varphi(F_t) = n \times Fr$$

$$S = L\varphi(s) = t \times s$$

$L\varphi|_{S\mathbb{R}}$ is ss $\Rightarrow S$ normalizes a Borel pair

conj. $\Rightarrow S$ normalizes $(\check{B}, \check{T})$

$W_t^0 \curvearrowright \check{G}$ normalizes $(\check{B}, \check{T})$

S normalizes $(\check{B}, \check{T})$

$\Rightarrow t$ normalizes $(\check{B}, \check{T}) \Rightarrow t \in \check{T}$.

We can further assume that $t \in \check{T}^{s,0} \leftarrow$ max'l form

$$\check{T}^{s,0} \longrightarrow \check{T}_s = \check{T} \times s / \check{T} \quad \text{ord } s = m$$

$$h \in \check{T}, \quad h = l^m$$

$$l^{-s}(l) \dots s^{m-1}(l) \in \check{T}^s$$

$$|| \text{ in } \check{T}_s$$

$$l^m = h.$$

We can assume that $t \in \check{T}^{s,0}$. See n s.t. $F = n \times Fr$.

$F S F^{-1} = S^2 \Rightarrow F$ normalizes a Borel pair in $L_G(S)$
 $(\check{B}^s, \check{T}^{s,0})$

conj. $\Rightarrow F$ normalize $\check{T}^{s,0} \Rightarrow F$ normalize $(\check{G}(\check{T}^s) = \check{T}$

$$S \subset \check{T}, \quad C_{\check{h}}(S) = (\check{T}, U_{\check{h}} : \check{\alpha} |_{\check{T}} \neq 1)$$

In our case, $\check{\alpha} |_{\check{T}^S} = 1$, then $\check{\alpha} + s \cdot \check{\alpha} + \dots + s^{m-1} \cdot \check{\alpha} = 0$ in $X^*(\check{T})$

OTOH, s normalize $(\check{B}, \check{T}) \Rightarrow s(\check{\Phi}^+) = \check{\Phi}^+$

$$\check{\alpha} \in \check{\Phi}^+$$

$$\sum n_i \alpha_i, \quad n_i \geq 0$$

contradiction.

$\frac{H}{0}$

F normalize $\check{T}^S, \check{T}$

Fr normalize $\check{T}^S, \check{T}$

$\leadsto n$ normalize $\check{T}^S, \check{T}$

One more input $FSF^{-1} = S^q$

$$\leadsto n \cdot Fr \cdot t \cdot s \cdot Fr^{-1} \cdot n^{-1} = (t \cdot s)^q$$

||

$$n \cdot Fr(t) \cdot Fr \cdot s \cdot Fr^{-1} \cdot n^{-1} = t^q s^q$$

$$t \in \check{T}^S$$

$$\leadsto n \cdot Fr(t) \cdot n^{-1} = t^q \cdot s^q(n) \cdot n^{-1}$$

$$\begin{array}{c} \text{||} \\ Fr(t) \\ \cap \\ \check{T}^S \end{array} \quad \begin{array}{c} \overline{\cap} \\ \check{T}^S \end{array}$$

$$\Rightarrow s^q(n) \cdot n^{-1} \in \check{T}^{S,0}$$

$$N_s = \{ n \in N_{\check{h}}(\check{T}) : n \text{ normalizes } \check{T}^{S,0}, \quad s^q(n) n^{-1} \in \check{T}^{S,0} \}$$

$$A = \{ (t, n) \in \check{T}^{s,0} \times N_s : n \text{Fr}(t) n^{-1} = t^q \cdot s^2(n) \cdot n^{-1} \}$$

$\Rightarrow$ there is a surjection

$$A(L) \rightarrow Z^1(W_t^0, \check{h}) // \check{h} (L)$$

$$(t, n) \mapsto L\varphi$$

$$\varphi(\text{Fr}) = n$$

$$\varphi(s) = t.$$

$$A // \check{T}^{s,0} \rightarrow Z^1(W_t^0, \check{h}) // \check{h}$$

finite $\rightarrow \downarrow$

$$N_s \rtimes \text{Fr} // \check{T}^{s,0}$$

$$\longrightarrow$$

$$\check{h} \rtimes \text{Fr} // \check{h}$$

$\uparrow$
finite.